

Pair of Straight Lines

A pair of straight lines in two dimensions is represented by a single quadratic equation with two variables, provided a specific condition is met.

General equation of a pair of straight lines:

The general second-degree equation for a pair of straight lines is:

$$ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$$

Important Formula:

1. General equation of second degree is,

$$ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$$

2. General equation of second degree of pair of straight line passing through the origin is,

$$ax^2 + 2hxy + by^2 = 0$$

3. The condition general equation of second degree of pair of straight line

$$\Delta = abc + 2fgh - af^2 - bg^2 - ch^2 = 0$$

4. General equation of second degree of pair of straight line then coordinate of intersection point

$$\text{of line } \left(\frac{fh - bg}{ab - h^2}, \frac{gh - af}{ab - h^2} \right).$$

5. General equation of second degree of pair of straight line then the angle between the line

$$\theta = \tan^{-1} \left(\frac{2\sqrt{h^2 - ab}}{a + b} \right).$$

Art. 48. From the above discussions the general equation of the second degree.
 $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$ will represent

- (1) a pair of straight lines if the determinant

$$\Delta \equiv \begin{vmatrix} a & h & g \\ h & b & f \\ g & f & c \end{vmatrix} = 0$$

- (i) two parallel lines if $\Delta = 0, ab = h^2$

- (ii) two perpendicular lines if $\Delta = 0, a + b = 0$

- (2) a circle if $a = b, h = 0,$

- ~~(3) a parabola if $ab = h^2, \Delta \neq 0$~~

- ~~(4) an ellipse if $ab - h^2 > 0, \Delta \neq 0$~~

- (5) a hyperbola if $ab - h^2 < 0, \Delta \neq 0$

- (6) a rectangular hyperbola if $a + b = 0, ab - h^2 < 0, \Delta \neq 0$

Note : $\Delta = abc + 2fgh - af^2 - bg^2 - ch^2 =$

$$\Delta = \begin{vmatrix} a & h & g \\ h & b & f \\ g & f & c \end{vmatrix}$$

Pair of straight lines

Let us consider the eqⁿ. $ax^2 + 2hxy + by^2 = 0$, $b \neq 0$①

we have, $\left(\frac{y}{x}\right)^2 + 2\frac{h}{b}\frac{y}{x} + \frac{a}{b} = 0$... ②

Let m_1, m_2 be the roots of this homogeneous quadratic equation in $\frac{y}{x}$.

Sum of the roots, $m_1 + m_2 = -\frac{2h}{b}$

and product of the roots, $m_1 m_2 = \frac{a}{b}$

The eqⁿ ① must be equivalent to $\left(\frac{y}{x} - m_1\right)\left(\frac{y}{x} - m_2\right) = 0$

The two lines represented by ① are given

$$\frac{y}{x} - m_1 = 0 \quad \text{and} \quad \frac{y}{x} - m_2 = 0$$

ie. $y - m_1 x = 0$ and $y - m_2 x = 0$ which pass through the origin.

Thus the homogeneous quadratic eqⁿ. $ax^2 + 2hxy + by^2 = 0$ always represents a pair of straight lines, real or imaginary, through the origin.

Note: 1. These two straight lines are real and different if $h^2 > ab$.

2. real and coincident if $h^2 = ab$

3. imaginary if $h^2 < ab$.

Angle between the lines represented by the equation $ax^2 + 2hxy + by^2 = 0$

If θ be the angle between the straight lines $y - m_1 x = 0$, $y - m_2 x = 0$

then, $\tan \theta = \pm \frac{m_1 - m_2}{1 + m_1 m_2}$

$$= \pm \frac{2\sqrt{h^2 - ab}}{a + b}$$

where, θ , the angle between the lines represented by the equation

$$ax^2 + 2hxy + by^2 = 0$$

Note: 1. Condition of perpendicularity of lines

$$\theta = 90^\circ \Rightarrow \tan \theta = \infty \text{ then } a+b=0$$

2. Condition of parallelism of the lines

$$\theta = 0^\circ \Rightarrow \tan \theta = 0 \text{ then } h^2 = ab$$

3. Condition for coincidence of the lines

$$h^2 - ab = 0, f^2 - bc = 0, g^2 - ac = 0$$

Bisectors of the angles betⁿ the lines

Let (α, β) be the pts of intersection of the two lines represented by the equation $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$.

The eqⁿs of the bisectors are given by

$$\frac{(x-\alpha)^2 - (y-\beta)^2}{(a-b)} = \frac{(x-\alpha)(y-\beta)}{h}$$

Where

$$\alpha = \frac{hf - bg}{ab - h^2}$$

$$\beta = \frac{hg - af}{ab - h^2}$$

Exp1: Find the eqⁿs of lines represented by $3x^2 - 16xy + 5y^2 = 0$.

Ans. $x - 5y = 0$, $3x - y = 0$

Exp2: Prove that the eqⁿ. $3y^2 - 8xy - 3x^2 - 29x + 3y - 18 = 0$ represents two straight lines; find also their point of intersection and the angles between them.

Ans. $(\frac{3}{2}, \frac{5}{2})$, 90°

Exp3: Find the value of k so that the eqⁿ. $6x^2 - 7xy - 3y^2 + 16x - 2y + k = 0$ may represent pair of straight lines.

Ans. 8

Exp4: Show that the straight lines represented by $6x^2 - 5xy - 6y^2 + 14x + 5y + 4 = 0$ are perpendicular to each other and $x^2 + 6xy + 9y^2 + 4x + 12y - 5 = 0$ are parallel.

Art. 42. Lines Joining the origin to the intersection of a curve and a line.

Let the second degree equation representing the curve be

$$ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0 \quad \dots \quad (1)$$

and the first degree equation representing the line be

$$lx + my = n \quad \dots \quad \text{or} \quad \frac{lx}{n} + \frac{my}{n} = 1 \quad \dots \quad (2)$$

Let the straight line meet the curve (1) at P and Q. We are to find the equation of the pair of lines OP and OQ.

Now making (1) homogeneous with the help of (2) we get

$$ax^2 + 2hxy + by^2 + (gx + fy) \left(\frac{lx + my}{n} \right) + c \left(\frac{lx + my}{n} \right)^2 = 0$$

Which, on simplification, reduces to an equation of the form

$$Ax^2 + 2Hxy + By^2 = 0 \quad \dots \quad (4)$$

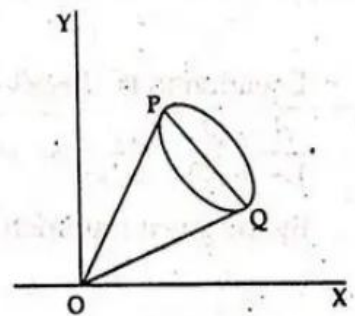


Fig. 13

✓ **Ex. 1.** Prove that the equation $x^2 + 6xy + 9y^2 + 4x + 12y - 5 = 0$ represents a pair of parallel lines. [R. U. 1997; D. U. 1986]

$$x^2 + 6xy + 9y^2 + 4x + 12y - 5 = 0$$

$$\text{or. } x^2 + x(6y + 4) + (9y^2 + 12y - 5) = 0 \text{ quadratic in } x$$

$$x = \frac{-(6y + 4) \pm \sqrt{(6y + 4)^2 - 4 \cdot 1 \cdot (9y^2 + 12y - 5)}}{2}$$

$$= (-6y - 4 \pm 6) / 2 = 3y - 2 \pm 3$$

$$x + 3y + 5 = 0 \text{ and } x + 3y - 1 = 0$$

Hence the given equation represents two straight line $x + 3y + 5 = 0$ and $x + 3y - 1 = 0$. As these two lines differ only in constant terms, so these form a pair of parallel straight lines. See

✓ Find the value of λ or k so that the following equations may represent pairs of straight lines.

$$8. \lambda x^2 + 4xy + y^2 - 4x - 2y - 3 = 0$$

Ans. 4 [D. U. 1960]

$$9. 2x^2 - y^2 + xy - 2x - 5y + k = 0$$

Ans. -4

$$10. 6x^2 + xy + ky^2 - 11x + 43y - 35 = 0$$

Ans. -12

$$11. x^2 - \lambda xy + 2y^2 + 3x - 5y + 2 = 0$$

Ans. $3, \frac{9}{2}$

$$12. kxy - 8x + 9y - 12 = 0$$

Ans. 6

$$13. 6x^2 + 2kxy + 12y^2 + 22x + 31y + 20 = 0$$

Ans. $\frac{171}{20}, \frac{17}{2}$ [D. U. 1962]

$$14. 6x^2 - 7xy + 16x - 3y^2 - 2y + k = 0$$

Ans. 8 [D.U. (Hons) 1962]

$$15. 12x^2 - 10xy + 2y^2 + 11x - 5y + \lambda = 0$$

Ans. $\lambda = 2$ [D. U. 1962]

36. Show that the lines joining the origin to the intersection of $x^2 + y^2 + 19x + 4y - 3 = 0$ and $3x + 4y = 1$ are at right angles. [R. H. 1988; D. U. 1961]

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